

Coding Discussion: Job Interview Practice...

Design and Analysis of Algorithms

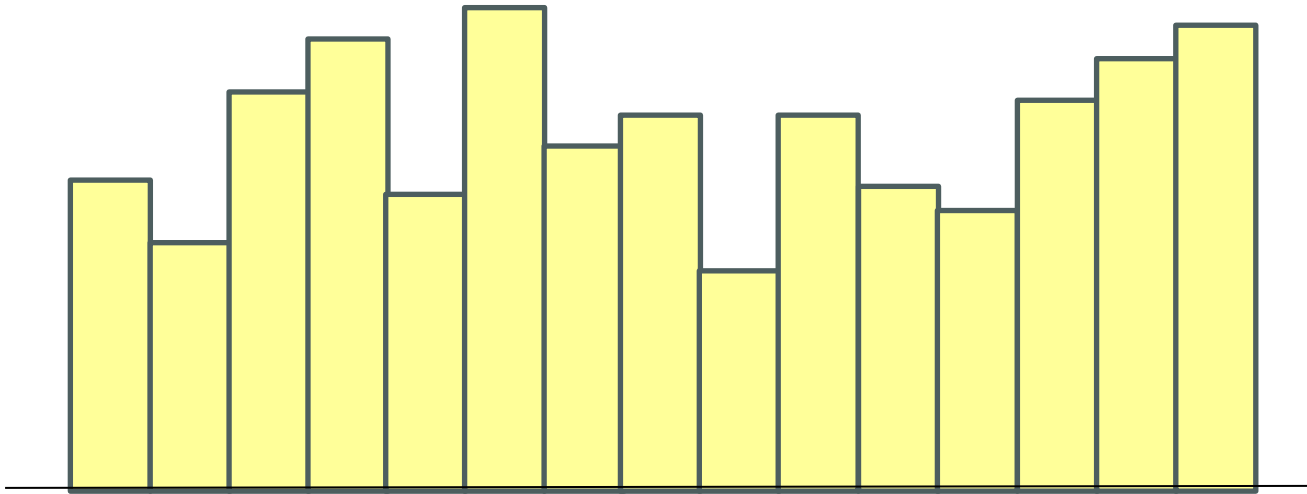
Brian C. Dean

**School of Computing
Clemson University**



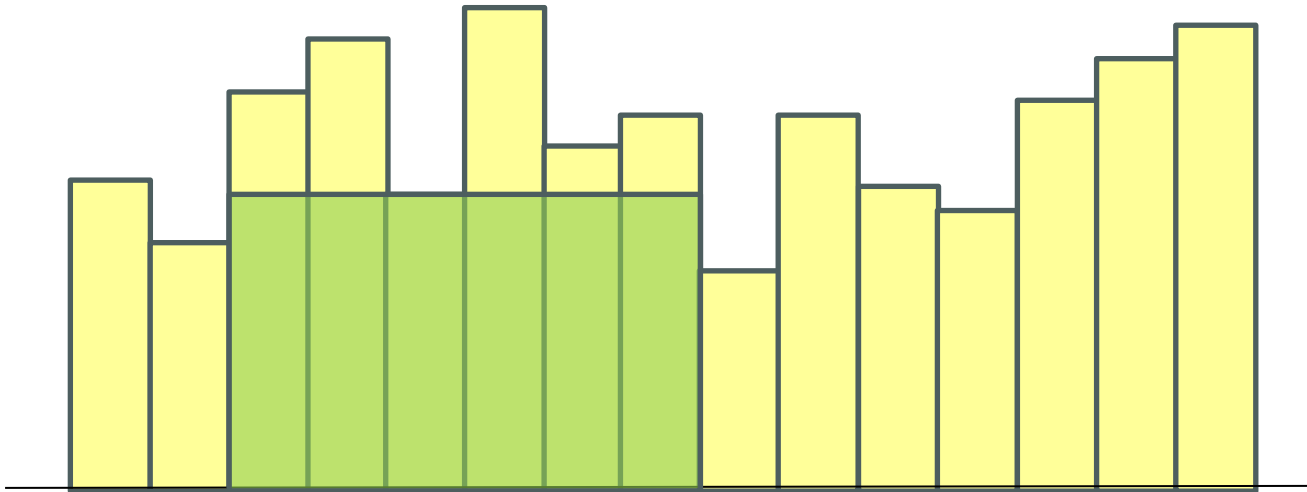
Today's Problem

Find the largest-area rectangle you can fit within a histogram (described by a sequence of n heights).



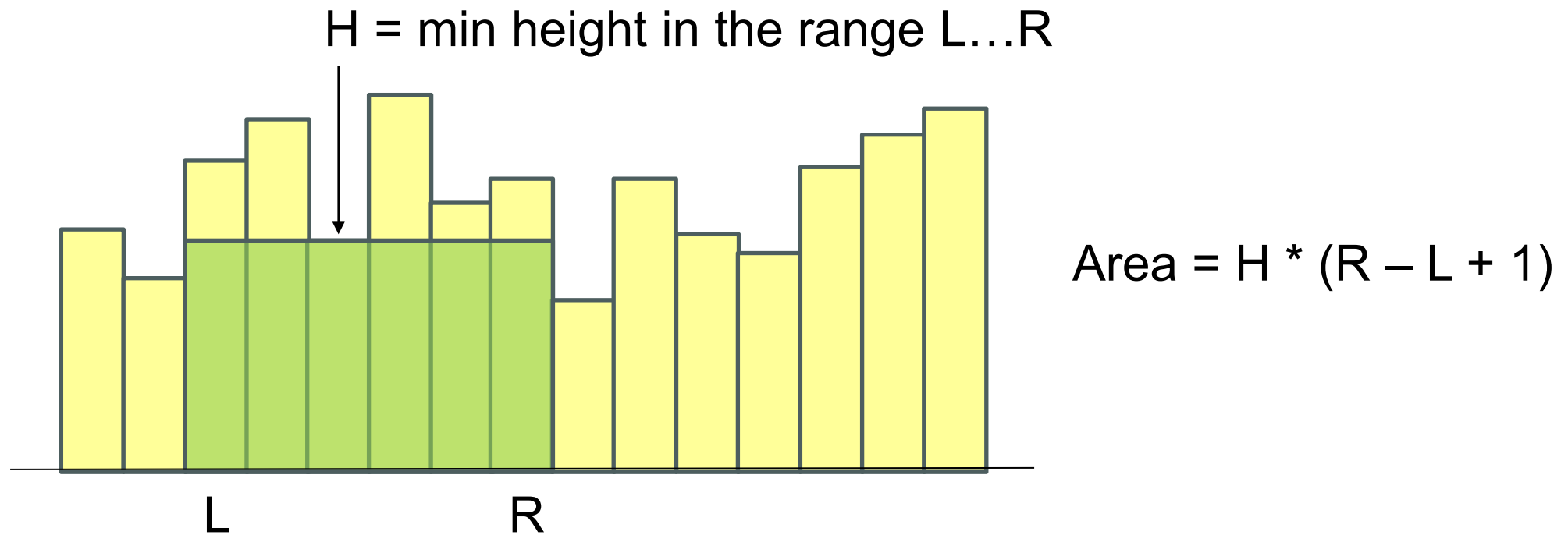
Today's Problem

Find the largest-area rectangle you can fit within a histogram (described by a sequence of n heights).



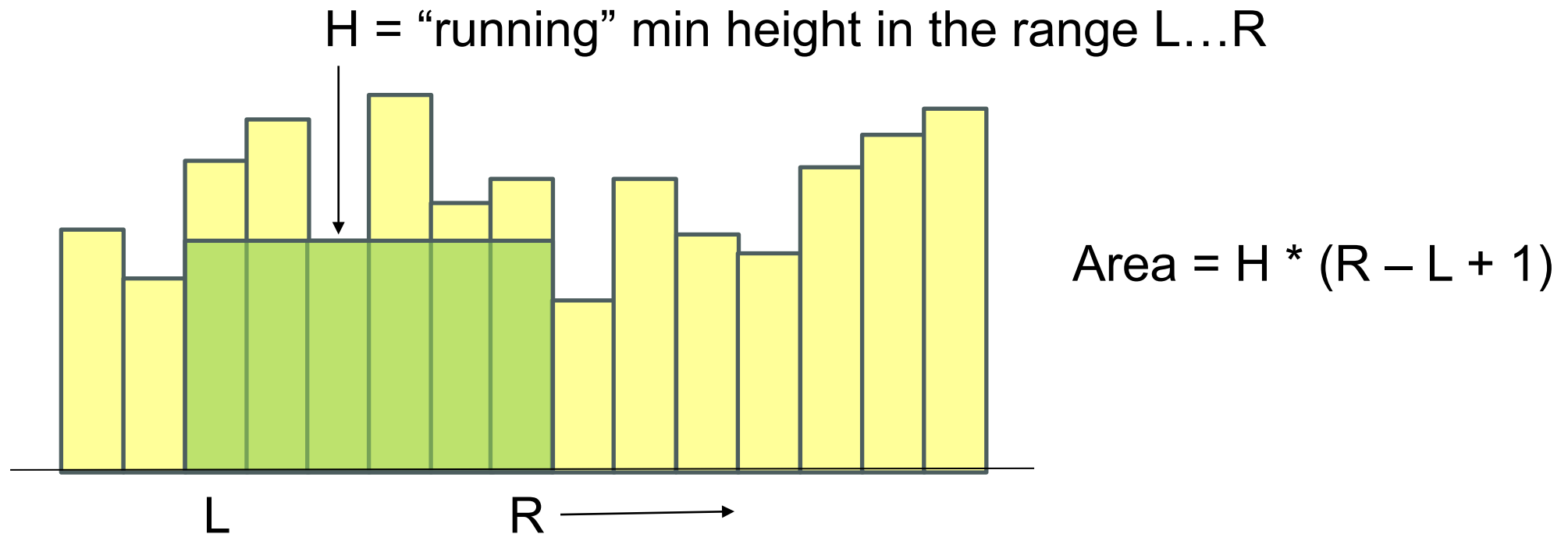
Today's Problem

Find the largest-area rectangle you can fit within a histogram (described by a sequence of n heights).



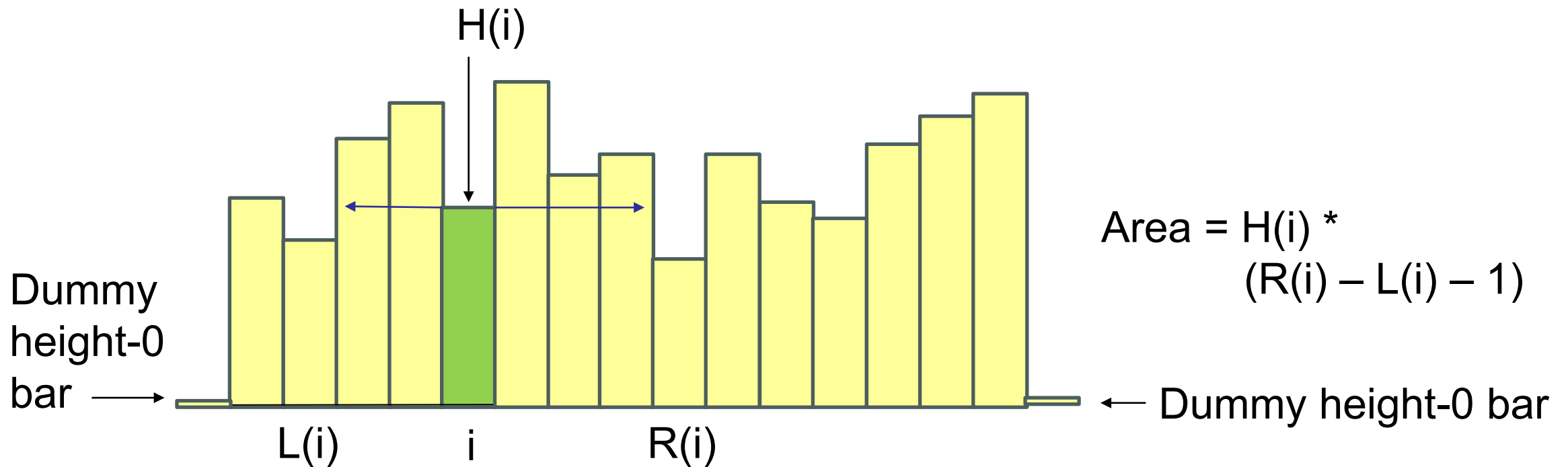
Today's Problem

Find the largest-area rectangle you can fit within a histogram (described by a sequence of n heights).



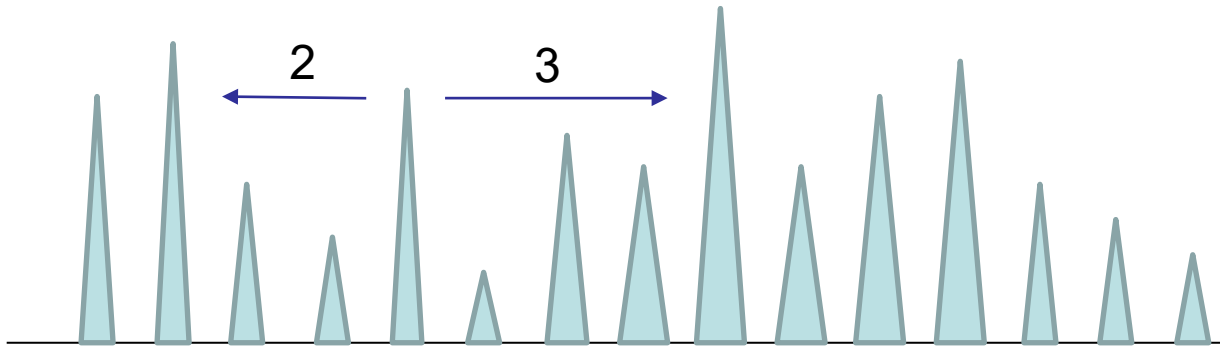
Today's Problem

New approach: for each rectangle i , what would be best rectangle if i were to define the top?



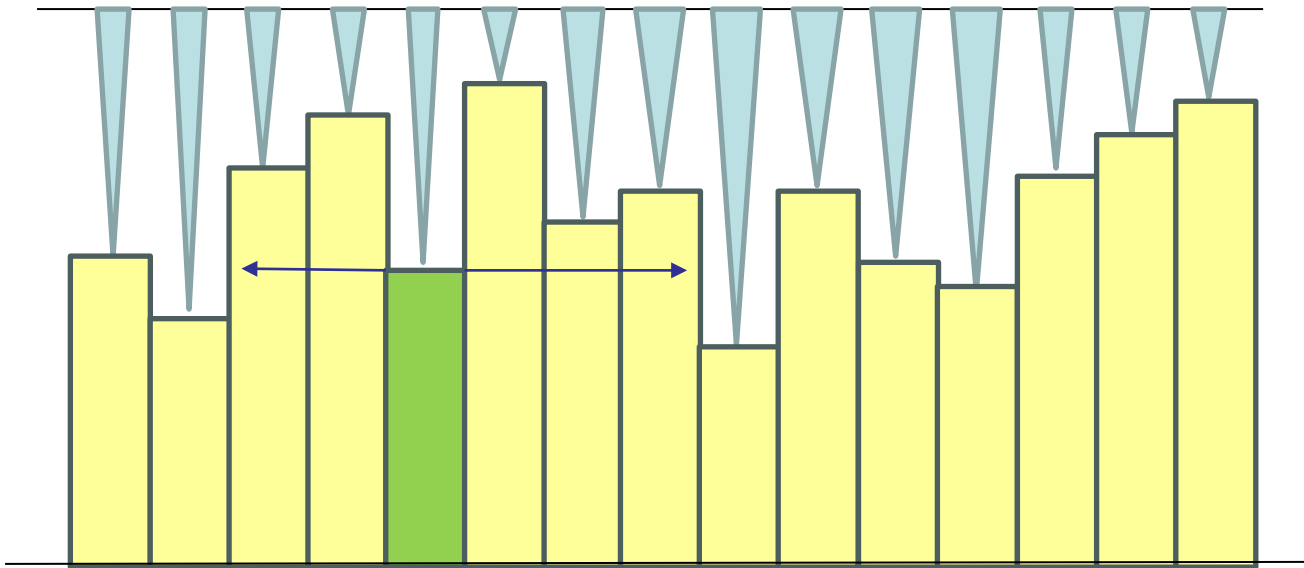
This Looks Maybe Familiar...

Remember the “domination radius” problem from our final mini-lecture on in-place matrix transposition?

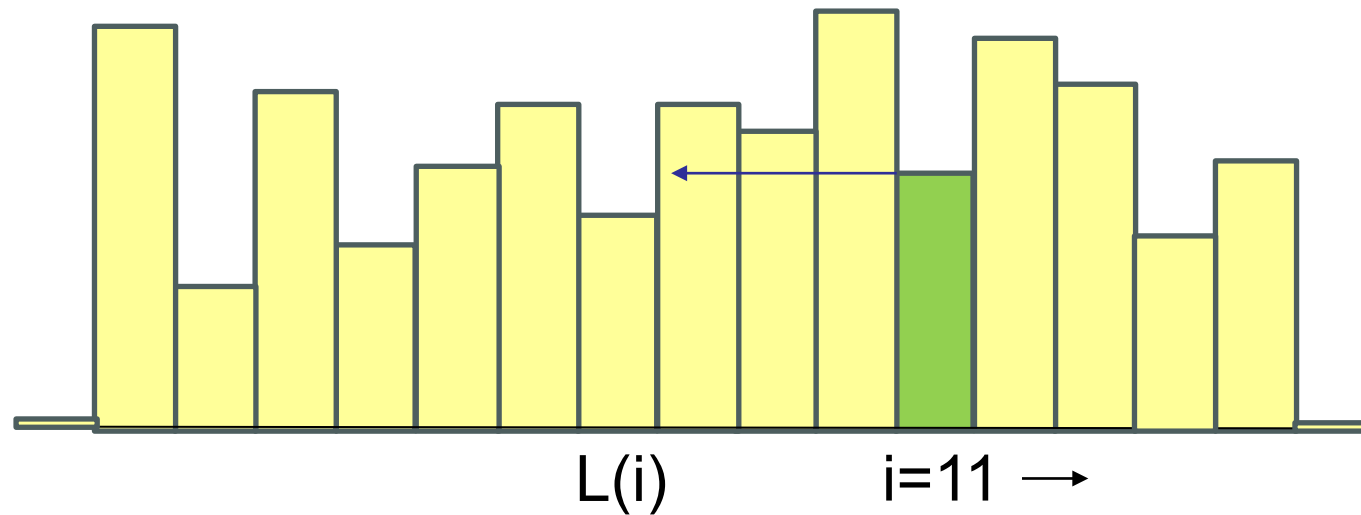
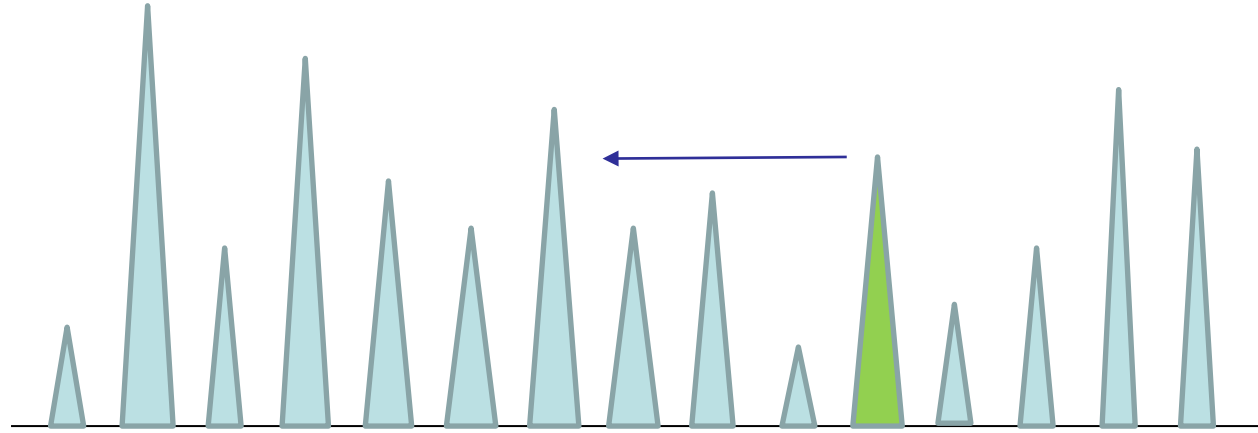


Well Isn't That a Nice Coincidence...

We can find the best rectangle by finding the left and right domination radii of each element.



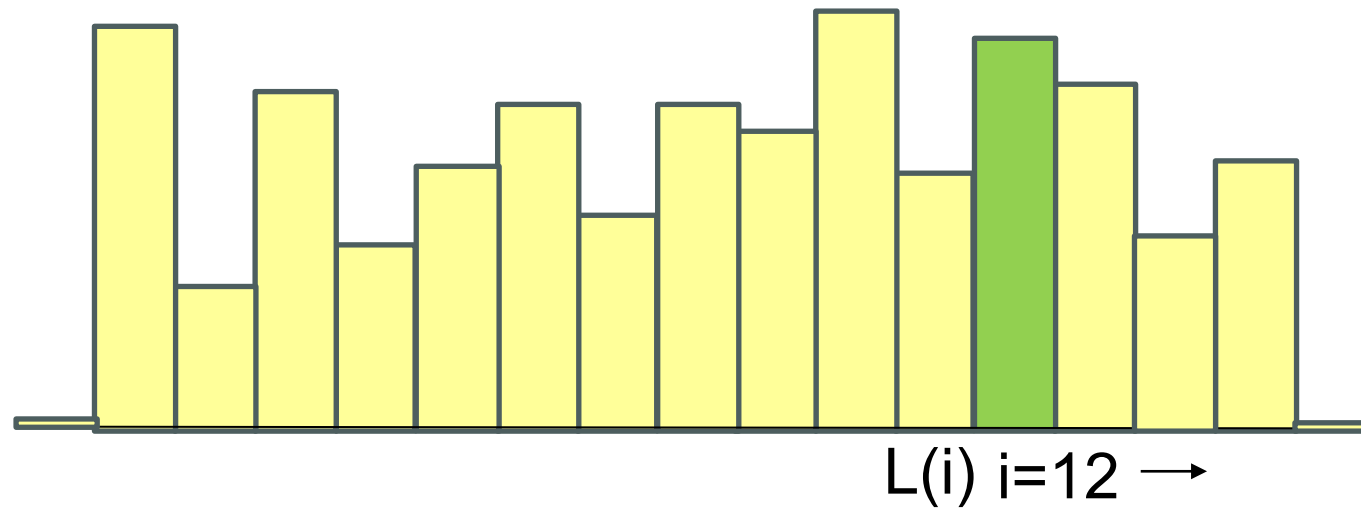
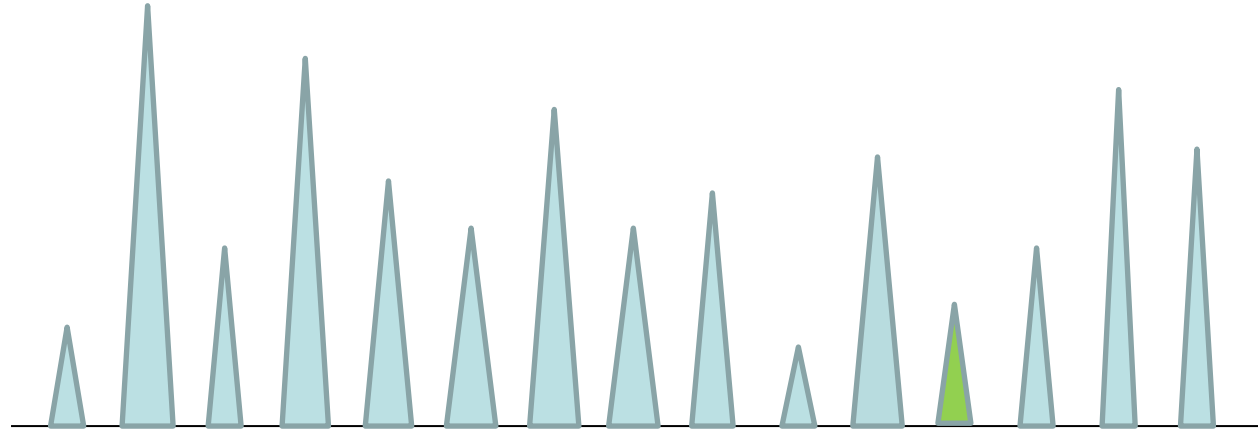
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

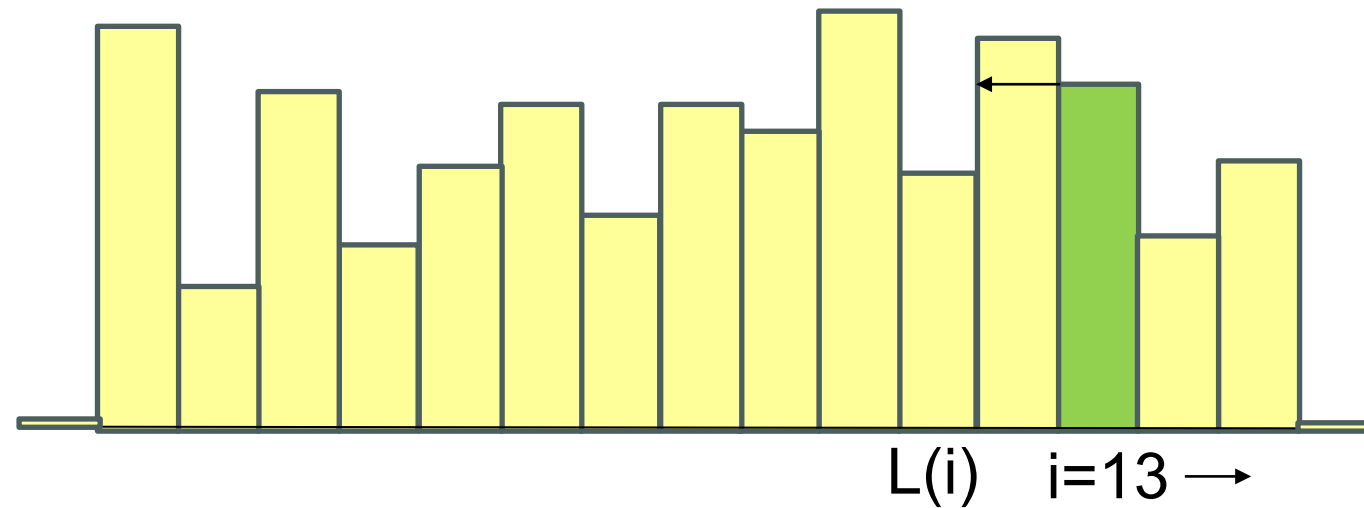
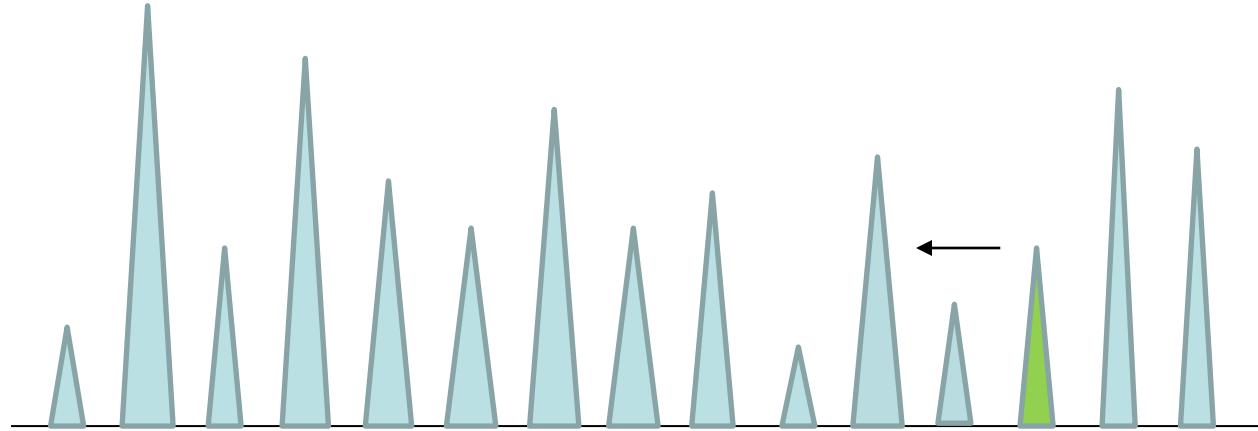
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10	11
---	---	---	---	---	---	---	---	---	----	----

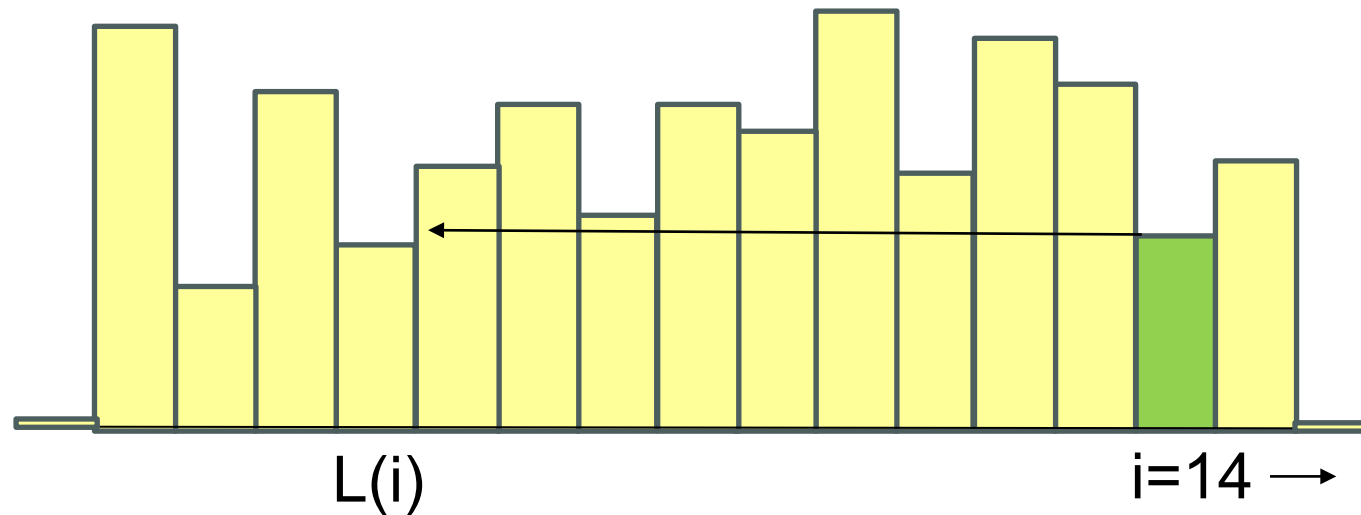
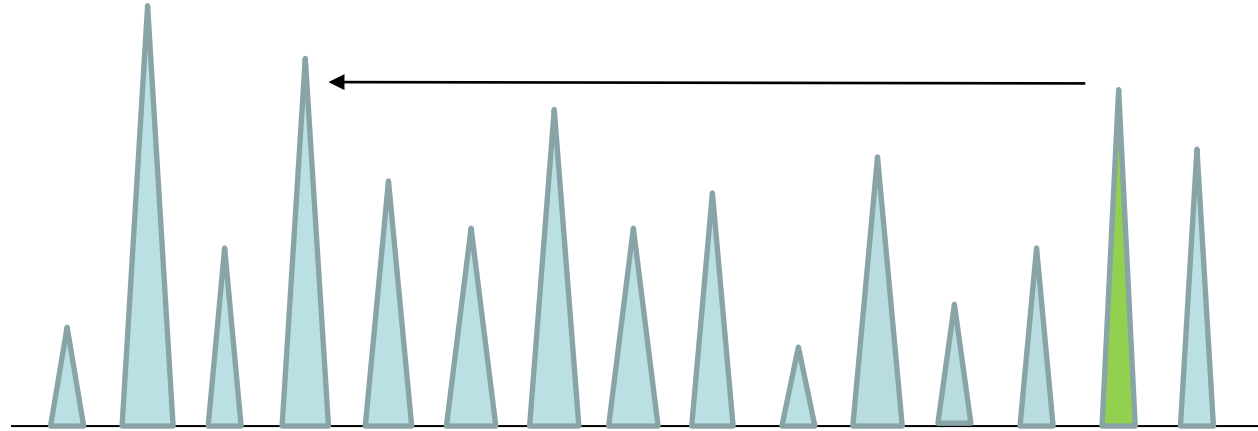
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	----	----	----

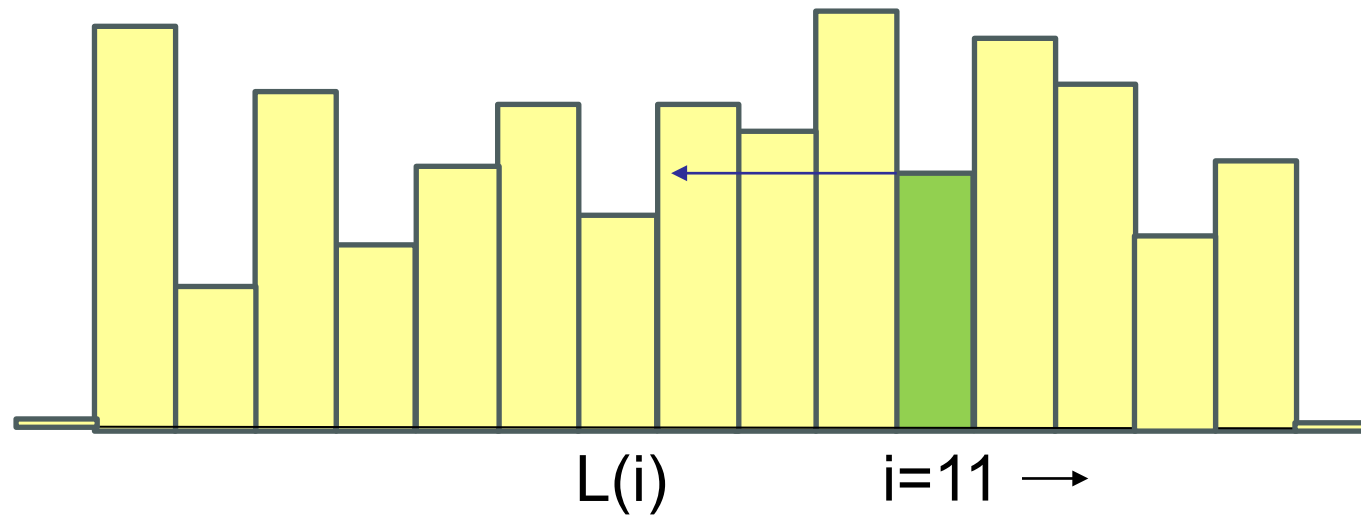
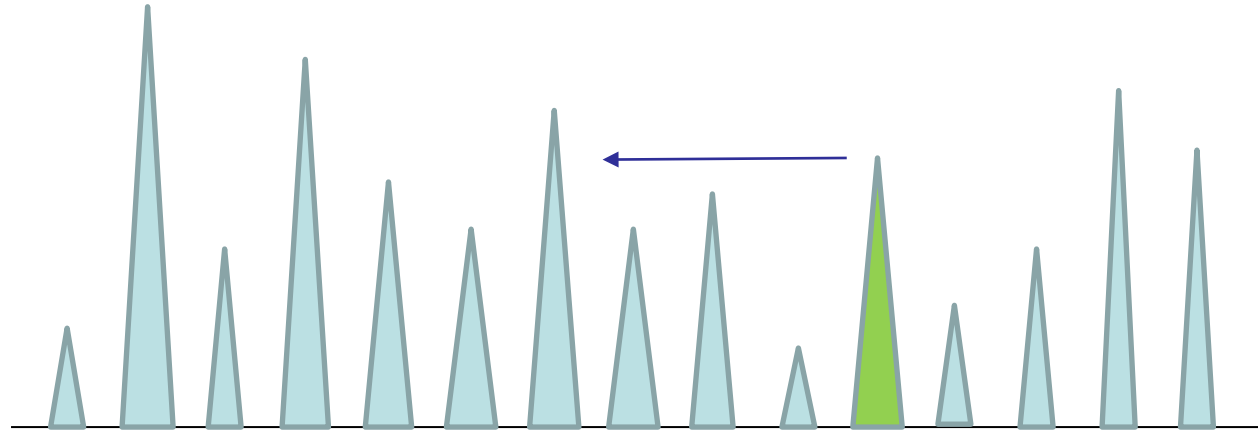
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10	11	12	13
---	---	---	---	---	---	---	---	---	----	----	----	----

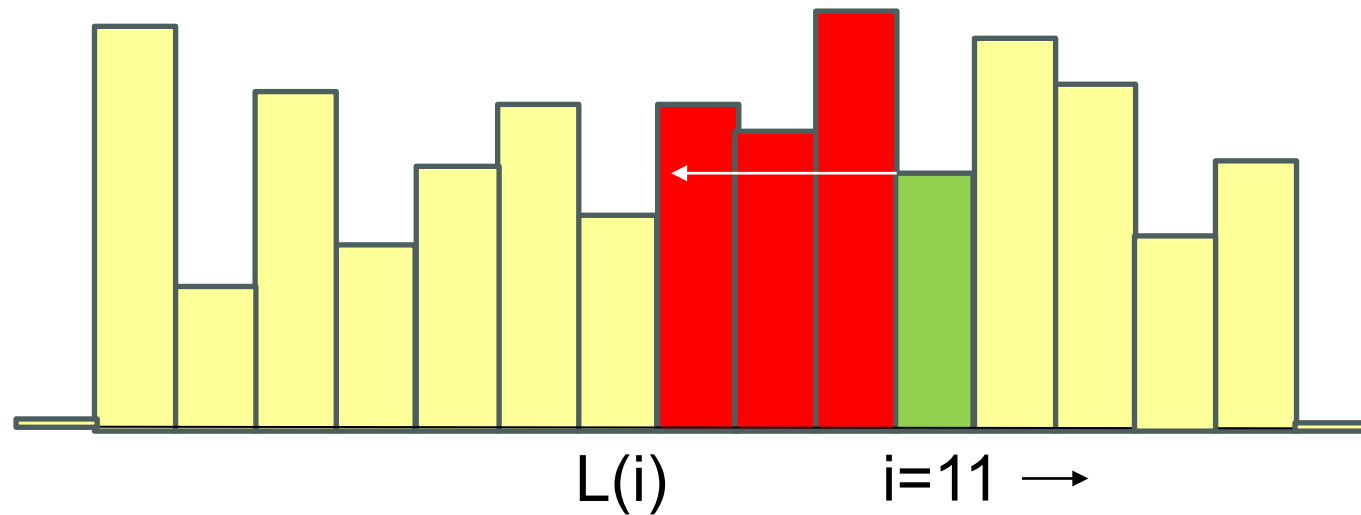
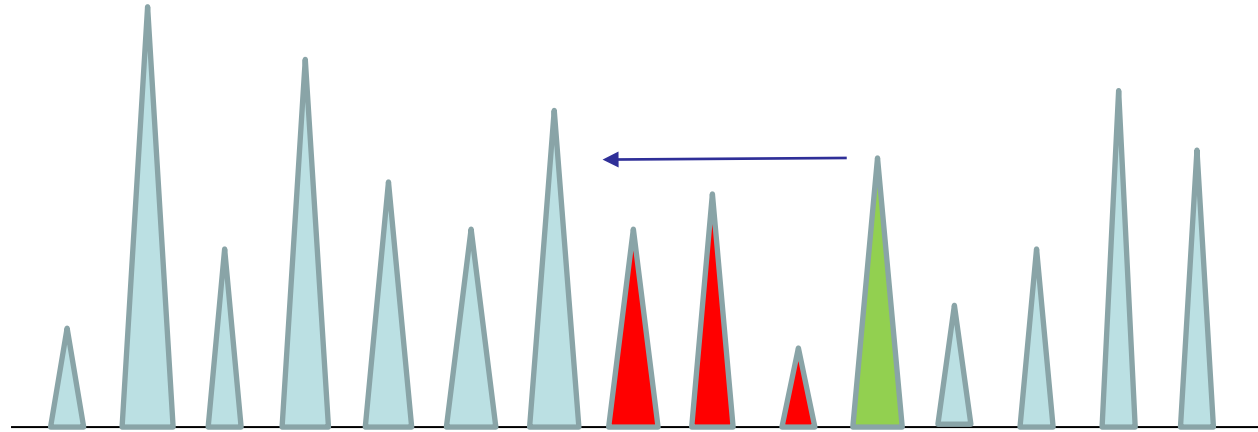
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

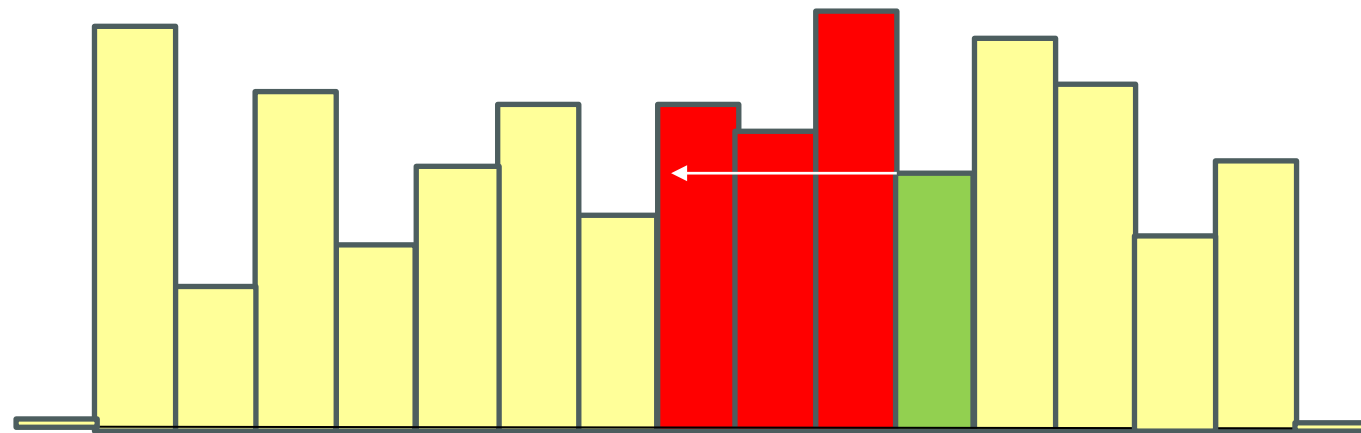
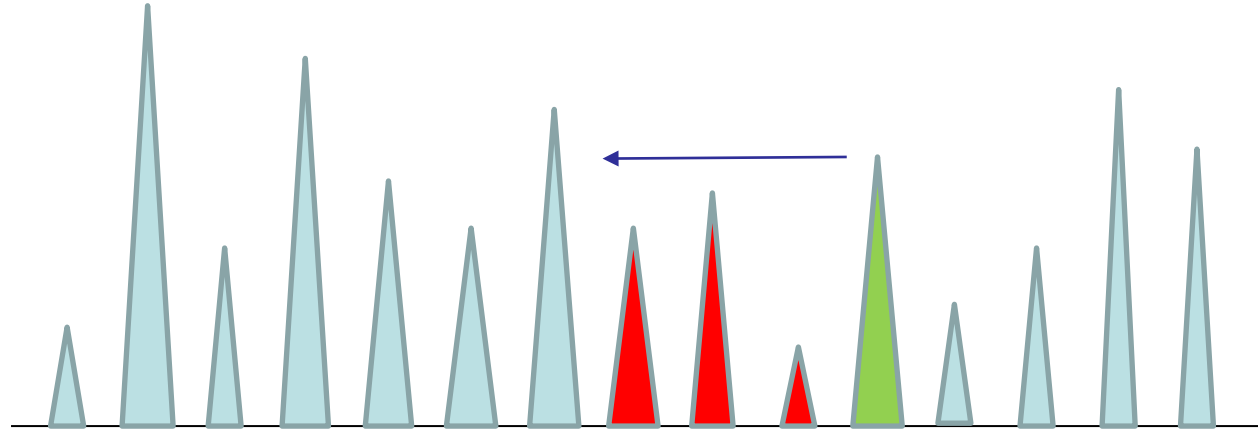
Compute $L(i)$ For Every i From $1 \dots N$



Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

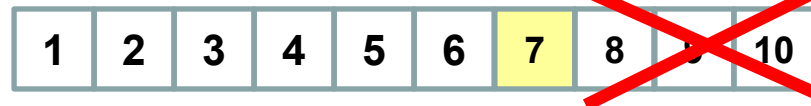
Compute $L(i)$ For Every i From $1 \dots N$



$L(i)$

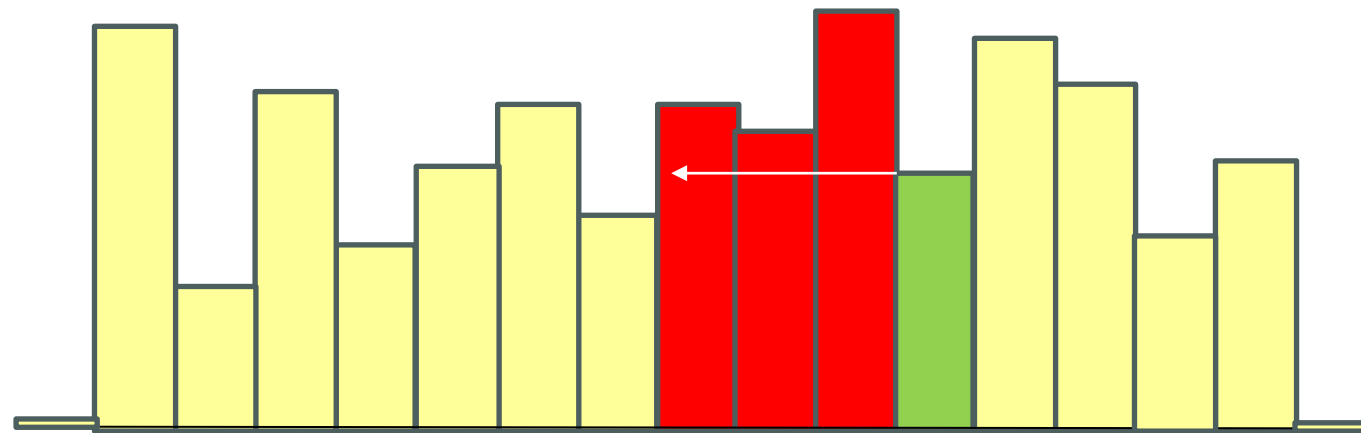
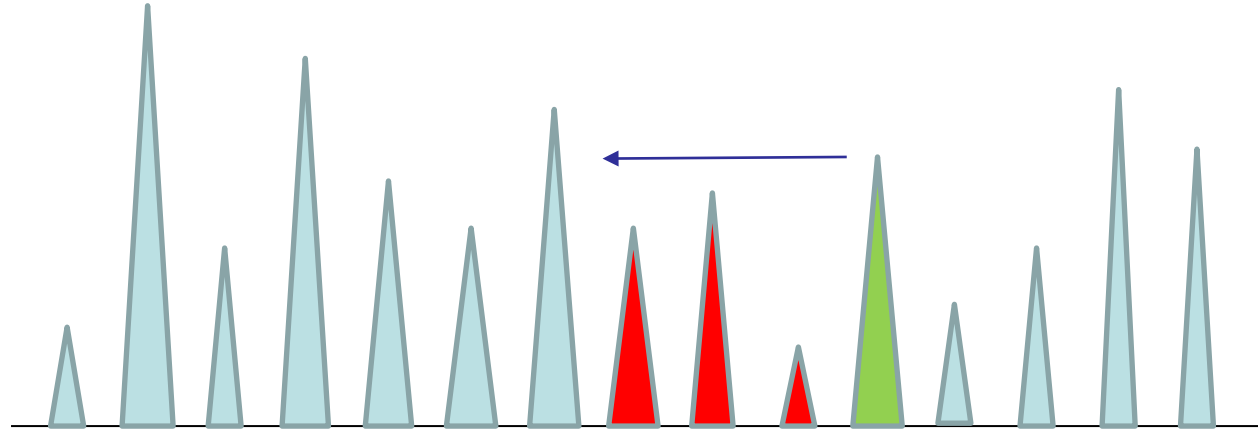
$i=11 \rightarrow$

Contenders for $L(i)$:



← Permanently remove!

Compute $L(i)$ For Every i From $1 \dots N$



$L(i)$ $i=11 \rightarrow$

Contenders for $L(i)$:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

 $\leftarrow$ Store as a stack

A red 'X' is drawn over the last three boxes (8, 9, 10) of the contender list.

Compute $L(i)$ For Every i From $1 \dots N$

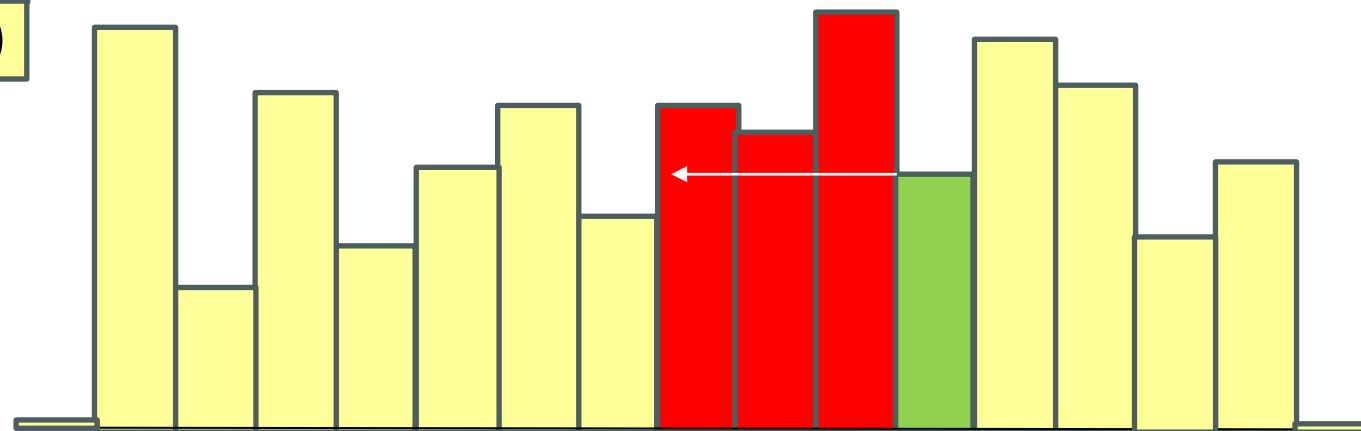
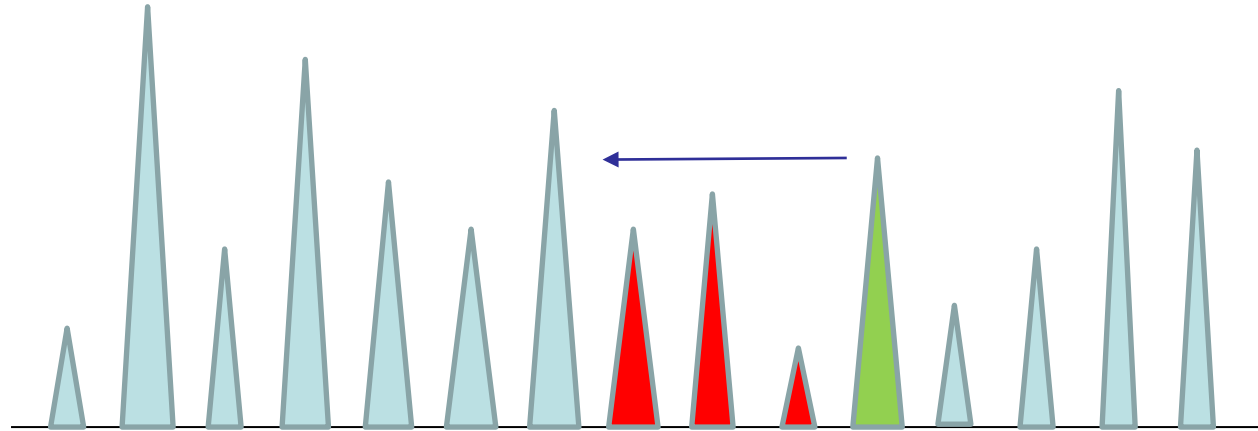
For each i :

1. Pop newly disqualified elements

(height $\leq H[i]$)

(height $\geq H[i]$)

2. $L[i] = \text{top}$
3. Push i



$L(i)$

$i=11 \rightarrow$

Contenders for $L(i)$:



$\leftarrow$ Store as a stack

Compute $L(i)$ For Every i From $1 \dots N$

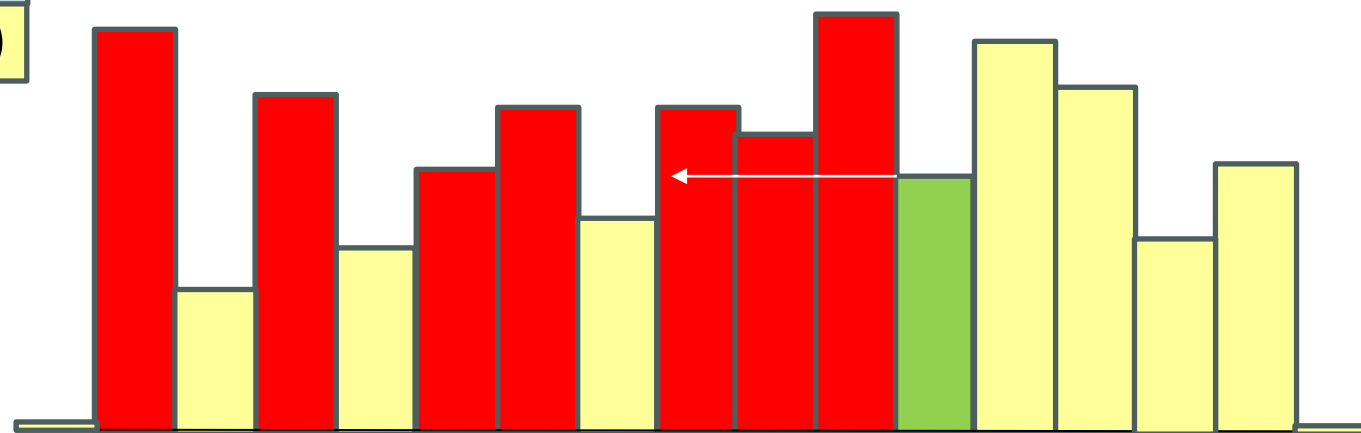
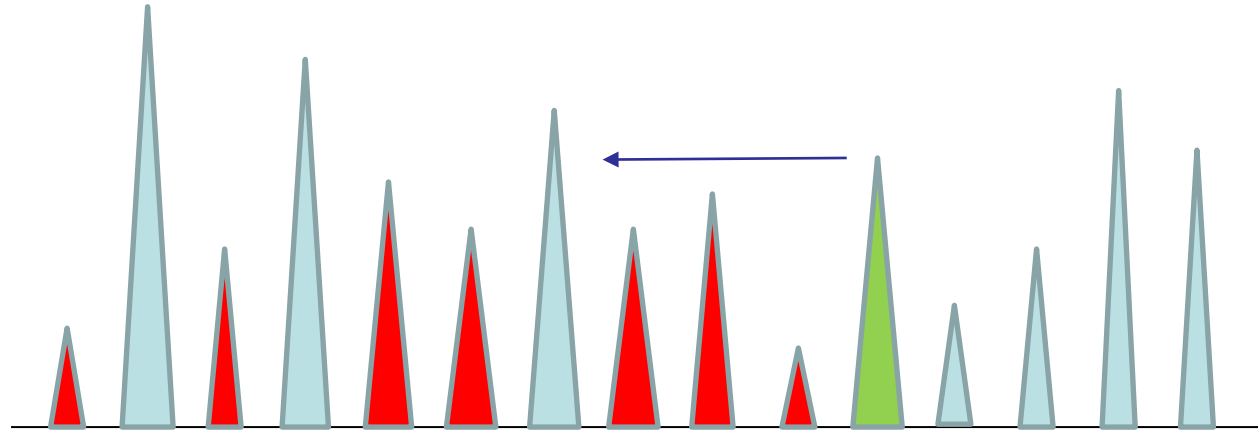
For each i :

1. Pop newly disqualified elements

(height $\leq H[i]$)

(height $\geq H[i]$)

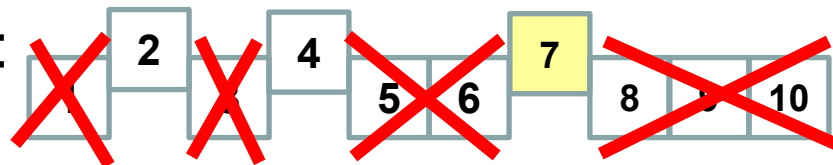
2. $L[i] = \text{top}$
3. Push i



$L(i)$

$i=11 \rightarrow$

Contenders for $L(i)$:



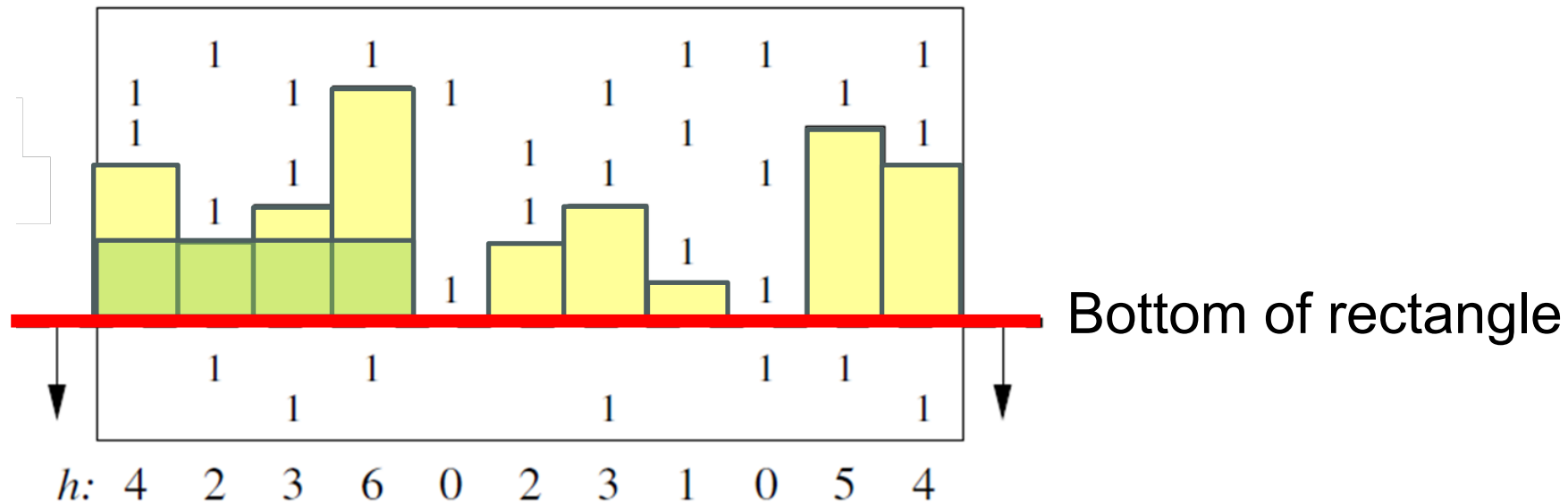
Store as a stack

One Notch More Challenging...

Find the largest-area rectangle of all 0s in an $N \times N$ binary matrix.

One Notch More Challenging...

Find the largest-area rectangle of all 0s in an $N \times N$ binary matrix.



Same problem, just repeated N times for different guesses of the bottom of the rectangle! $\Theta(N^2)$ running time.